CHAPTER 5

ESTIMATION OF PARAMETERS

CHAPTER ESSENCE AT A GLANCE:
Master point estimation vs interval estimation, four criteria of an ideal estimator (unbiasedness, consistency, efficiency, sufficiency), and confidence interval construction.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Point vs Interval Estimation:** Point estimation provides a single numerical value as an estimate of an unknown parameter; Interval estimation constructs a range with specified confidence probability 1 - alpha.

• **Criteria of a Good Estimator:** An ideal estimator must satisfy four criteria: Unbiasedness [E(T) = theta], Consistency [plim T = theta], Efficiency [minimum variance], and Sufficiency [utilizes all sample info].

• **Sample Mean and Variance Properties:** Sample mean x_{bar} is an unbiased and consistent estimator of population mean mu; Sample variance s^2 with divisor n-1 is unbiased for population variance $sigma^2$.

• **Confidence Interval Construction:** Confidence limits = Point Estimate +- (Critical Value * Standard Error); 95 percent confidence interval uses Z = 1.96; 99 percent uses Z = 2.58.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Unbiasedness Condition} \iff E(\hat{\theta}) = \theta \quad \text{for all } \theta \in \Theta$$

Zero Long-Run Systematic Bias Requirement

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} \rightarrow E(s^2) = \sigma^2 \quad [\text{Unbiased Variance Estimator}]$$

Bessels Correction for Sample Variance Unbiasedness

$$\text{Confidence Interval for } \mu \iff \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Two-Sided Symmetric Population Mean Confidence Bounds

$$Z_{0.05} = 1.96 \quad (95\%), \quad Z_{0.01} = 2.58 \quad (99\%), \quad Z_{0.10} = 1.645 \quad (90\%)$$

Standard Gaussian Critical Multiplier Constants

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Sample variance with divisor n [sum (xi - x_{bar})^2 divided by n] is BIASED; dividing by n - 1 (Bessel's correction) makes it strictly unbiased: E(s^2) = $sigma^2$.

• **Alert:** A 95 percent confidence interval does NOT mean there is a 95 percent probability that the true parameter falls in that specific calculated interval; it means 95 percent of similarly constructed intervals contain the parameter.

• **Alert:** Efficiency is relative; between two unbiased estimators, the one with smaller variance is more efficient.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Point Estimation and Interval Estimation. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Point Estimation:** Calculates a single specific numerical value from sample data to serve as the best estimate of the unknown population parameter (e.g. Sample mean x_{bar} = 50). [1 Mark]

• **Interval Estimation:** Constructs a range or interval of numbers [$theta_L$, $theta_U$] calculated from the sample that is expected to contain the true parameter with a specified confidence level (e.g. 48 to 52 with 95 percent confidence). [1 Mark]

2. Explain the four essential criteria of a good estimator. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Unbiasedness:** The expected value of the estimator T equals the true parameter theta [E(T) = theta] across repeated sampling. [1 Mark]

• **2. Consistency:** As sample size n increases indefinitely, the estimator converges in probability to the true parameter value theta. [1 Mark]

• **3. Efficiency:** Among all competing unbiased estimators, the efficient estimator possesses the minimum sampling variance. [1 Mark]

• **4. Sufficiency:** The estimator utilizes all the relevant information contained in the sample data concerning the parameter. [1 Mark]

3. A random sample of 64 bulb lifetimes gave a mean of 1,200 hours. If the population standard deviation is 80 hours, construct a 95 percent confidence interval for the true population mean lifetime. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Given Parameters:** n = 64, x_{bar} = 1,200, σ = 80. For 95 percent confidence, Z critical value = 1.96. [1 Mark]

• **Standard Error:** SE = sigma divided by square root of n = 80 divided by square root of 64 = 80 divided by 8 = 10 hours. [1 Mark]

• **Confidence Limits:** x_{bar} +- 1.96 * SE = 1,200 +- (1.96 × 10) = 1,200 +- 19.6. [1 Mark]

• **Result:** Lower limit = 1,200 - 19.6 = 1,180.4 hours; Upper limit = 1,200 + 19.6 = 1,219.4 hours. The 95 percent confidence interval is [1180.4, 1219.4] hours. [1 Mark]

4. In a sample of 400 voters, 240 favored a specific candidate. Construct a 95 percent confidence interval for the true proportion of voters favoring the candidate. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Sample Proportion:** p = 240 divided by 400 = 0.60; q = 1 - 0.60 = 0.40; n = 400. [1 Mark]

• **Standard Error:** SE(p) = square root of [(p * q) divided by n] = square root of [(0.60 * 0.40) divided by 400] = square root of [0.24 divided by 400] = square root of 0.0006 = 0.0245. [1 Mark]

• **Confidence Interval:** p +- 1.96 * SE = 0.60 +- 1.96 * 0.0245 = 0.60 +- 0.048 = [0.552, 0.648] (or 55.2 percent to 64.8 percent). [1 Mark]

5. Why is the divisor n - 1 used instead of n when computing sample variance as an estimator of population variance? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Explanation:** Dividing by n introduces a systematic downward bias: E[sum(xi - x_{bar})^2 divided by n] = [(n - 1) divided by n] * σ^2. [1 Mark]

• **Bessel Correction:** Using divisor n - 1 corrects for this bias, ensuring that the estimator is strictly unbiased: E(s^2) = σ^2. [1 Mark]

6. What happens to the width of a confidence interval when: (a) Sample size n increases, (b) Confidence level increases from 95 percent to 99 percent? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) Sample Size Increases:** Standard error decreases (SE propto 1 divided by sqrt(n)), making the confidence interval narrower and more precise. [1 Mark]

• **(b) Confidence Level Increases:** Critical value Z increases (from 1.96 to 2.58), making the confidence interval wider to ensure higher certainty. [1 Mark]